

Optimization & Multi-Collinearity

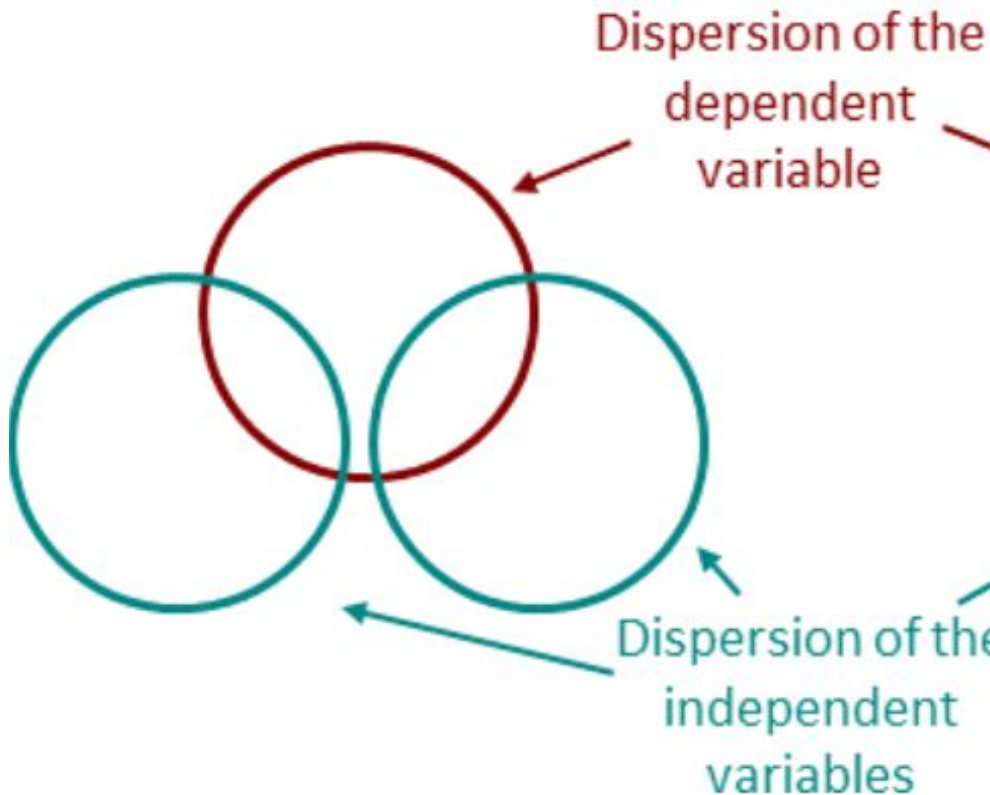
Multi-Collinearity – Concept

- • Multi-collinearity occurs when two or more independent variables are highly correlated.
- • Multicollinearity in linear regression occurs when independent variables are highly correlated, making it difficult to isolate the individual effect of each variable on the dependent variable, leading to unstable and unreliable regression coefficient estimates.

Multi-Collinearity – Concept

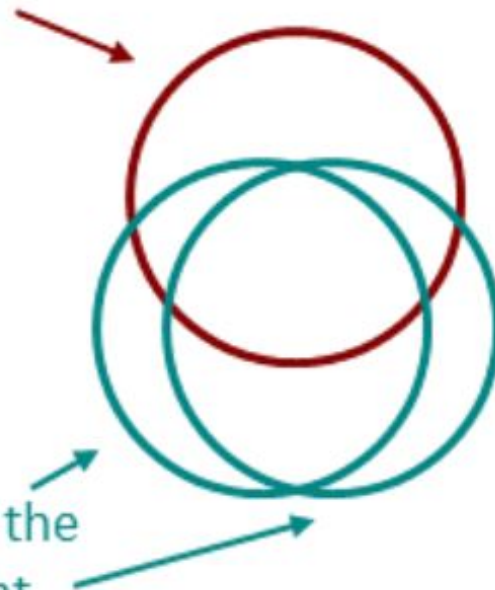
No multicollinearity

No correlation of the independent variables



Multicollinearity

High correlation of the independent variables



Multi-Collinearity – Concept

- **Why it's a problem:**
- **Unstable coefficients:** The estimated regression coefficients become unstable and difficult to interpret, as the effect of one variable can be influenced by the presence of other correlated variables.
- **Inflated standard errors:** The standard errors of the coefficients increase, leading to wider confidence intervals and potentially making statistically insignificant variables appear significant.
- **Difficulty in isolating effects:** It becomes challenging to determine the individual effect of each independent variable on the dependent variable because the correlated variables are intertwined.
- **Detect with correlation matrix, Variance Inflation Factor (VIF), condition index.**

Multi-Collinearity – SLR vs MLR

- • SLR: Only one independent variable; no multicollinearity.
- • MLR: Multiple predictors; high correlation among predictors introduces multicollinearity.
- Example: Predicting price based on area, no. of rooms, and total sqft – rooms and sqft may correlate.

Detecting Multi-Collinearity – VIF

- • Variance Inflation Factor (VIF):

$$VIF = 1 / (1 - R^2)$$

- • If $VIF > 5$ or $10 \rightarrow$ strong multicollinearity exists.
- Remedies: Drop variables, use PCA or Ridge Regression.

Optimization in Data Analytics – Example

- • Goal: Minimize loss in linear regression.
- • Loss Function: $MSE = (1/n) \sum (y_i - \hat{y}_i)^2$
- • Use gradient descent to find optimal weights (coefficients).
- • Optimization improves model prediction accuracy.

Unconstrained Optimization – Concept

- It is the process of finding the **minimum** or **maximum** of a function **without any restrictions** (constraints) on the variables.
- • Minimize $f(x)$, without any constraints
- **General Form:**
- Minimize or Maximize $f(x)$
- Where $f(x)$ is a function of one or more variables and **no constraints** are applied.

Unconstrained Optimization – Concept

- **Steps in Unconstrained Optimization (Single Variable)**
- **Compute the derivative:** $f'(x)$
- **Set it to zero** to find critical points: $f'(x)=0$
- **Use second derivative $f''(x)$:**
 - If $f''(x)>0$, it's a **minimum**
 - If $f''(x)<0$, it's a **maximum**
- **Example:** Minimize $f(x) = x^2$
- **Solution:** $x = 0$, ie $f(x)$ is minimum at $x=0$

Constrained Optimization – Concept

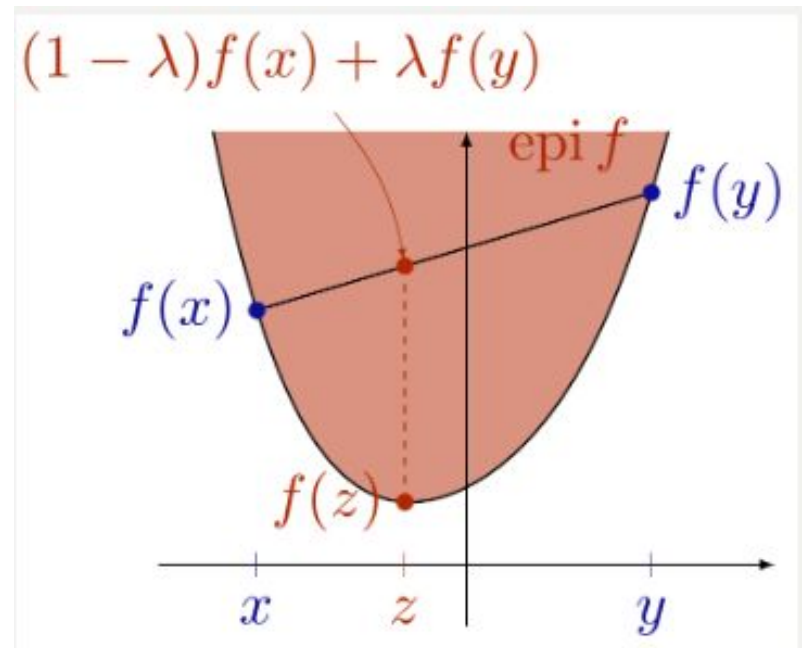
- Variables must satisfy equality or inequality constraints.
- Example: Minimize $f(x, y) = x^2 + y^2$, subject to $x + y = 1$
- Requires Lagrangian or KKT conditions.

Real-life Examples – Constraints

- • Portfolio Optimization: Maximize return with budget and risk limits.
- • Production Planning: Maximize output with raw material constraints.
- • ML Regularization: Constrain weights to prevent overfitting.

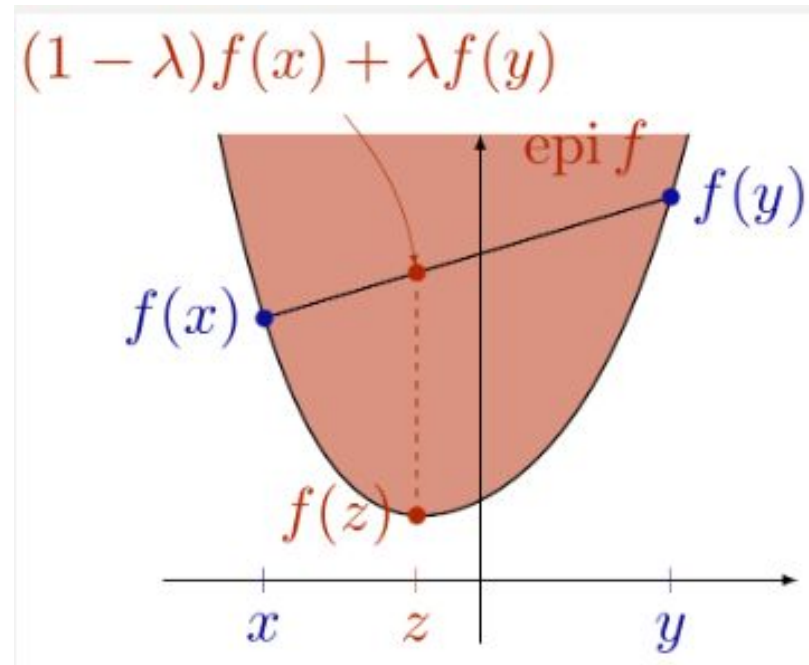
Convex Functions – Introduction

- $f(x)$ is convex if line segment between any two points on graph lies above graph.
- If you pick two points on the graph of a convex function, the line connecting them will not dip below the curve itself.



Convex Functions – Introduction

- $f(x)$ is convex if line segment between any two points on graph lies above graph.
- Convex if
- $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) \quad \forall \lambda \in [0,1]$



Convex Functions – Properties

- • Local minimum is also global minimum.
- Convex functions often appear in optimization problems, as they have nice properties that make finding their minima relatively easy compared to other types of functions.
- • First-order condition: $f(y) \geq f(x) + \nabla f(x)^T(y - x)$
- • Second-order: Hessian $H(x) \geq 0$

Importance of Convexity

- • Guarantees convergence to global minima. This means that optimization algorithms don't get "stuck" in suboptimal solutions, making it straightforward to identify the best solution.
- • Convex optimization problems can often be solved efficiently using techniques like gradient descent.

Importance of Convexity

- For example, in logistics, minimizing the cost of delivery routes involves solving a convex optimization problem. Similarly, in machine learning, minimizing a loss function like mean squared error to train a regression model is also an example of convex optimization.
- • Helps in designing reliable and efficient optimization algorithms.

Gradient Descent – Example

- • Minimize $f(x) = (x-3)^2$
- • $\nabla f(x) = 2(x-3)$
- • Update: $x \leftarrow x - \alpha \nabla f(x)$
- • Start at $x = 0$, $\alpha = 0.1$
- Iter1: $x = 0.6$, Iter2: $x = 1.08$...
- • Converges to $x = 3$

Subgradient Descent – Example

- • $f(x) = |x|$ is not differentiable at $x = 0$.
- • Subgradient $g = -1$ if $x < 0$, 1 if $x > 0$, $[-1, 1]$ if $x = 0$
- • $x \leftarrow x - \alpha g$
- • Useful in L1 regularization problems.

Newton's Method – Example

- • $f(x) = x^2$, $\nabla f(x) = 2x$, $H(x) = 2$
- • Update: $x \leftarrow x - \nabla f(x)/H(x) = 0$
- • Converges in one step if function is quadratic.
- • Faster than gradient descent near optimum.

Lagrangian Method – Example

- • Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 1$
- • $L(x, y, \lambda) = x^2 + y^2 + \lambda(1 - x - y)$
- • $\nabla L = 0 \Rightarrow x = y = 0.5$

Lagrangian in ML

- • Regularization: Add $\lambda ||w||^2$ to cost to penalize complexity.
- • Support Vector Machines: Constrained optimization using Lagrange duality.

Summary

- • Multi-collinearity affects model interpretation.
- • Gradient and subgradient methods are iterative optimizers.
- • Newton's method is faster but uses Hessian.
- • Lagrangian method handles constraints effectively.
- All techniques are fundamental for advanced data science and AI.

Question and Answer

- 1. When do we prefer subgradient over gradient descent?
- We prefer subgradient methods when the objective function is not differentiable. For example, in optimization problems involving absolute values or hinge loss, gradient descent is not applicable because the gradient doesn't exist everywhere. Subgradients provide a way to still optimize such functions.

Question and Answer

- 2. What does a Lagrange multiplier represent?
- A Lagrange multiplier represents how much the objective function would increase or decrease if the constraint boundary were relaxed slightly. It gives insight into the sensitivity of the objective function to constraints.

Question and Answer

- 3. Why is convexity important in optimization?
- Convexity ensures that any local minimum is also a global minimum. This property makes optimization more reliable and easier because methods like gradient descent will converge to the best solution.

Question and Answer

- 4. Compare Newton's method and gradient descent.
- Gradient Descent uses first-order derivative information and is simpler but slower. Newton's Method uses second-order derivatives (Hessian) and converges faster near the minimum, but it's computationally more expensive.

Question and Answer

- 5. How does multi-collinearity affect predictions?
- Multi-collinearity inflates the variance of coefficient estimates in regression models. This makes predictions unstable and sensitive to small changes in data, reducing model interpretability and reliability.